

Experiment	Material	Dopant	%wt.	IBDs	Vol. (L)	N	$\overline{c\psi}$	$\overline{c\psi}/N \times 10^6$
CHOOZ-type	DC LOS	Gd	0.1	10 000	785 000	9198	0.030	3.22
NuLat (3^3)	EJ-Base	^6Li	1.5	10 000	15.6	4934 [†]	0.224 [†]	45.40 [†]
NuLat (5^3)	EJ-Base	^6Li	1.5	10 000	422.	3931	0.256	65.20
3D Chkbd.	EJ-Base	^6Li	1.5	10 000	107.	4206	0.194	46.10
SANTA	EJ-Base	B	0/5	10 000	24.0	247	0.360 [‡]	1460.00 [‡]
SANDD	EJ-Base	^6Li	1.5	10 000	0.64	1860	0.297 [‡]	160.00 [‡]
2D Chkbd.	EJ-Base	^6Li	1.5	10 000	5.12	7752	0.266	34.30

Table 11.1: **Summary Table of New Simulation Results**

- “Vol.” is the detector’s active target volume; SANTA’s capture plane and the inert checkerboard cells are not included.
 - $\overline{c\psi} \equiv \langle \cos(\psi) \rangle$, the mean of $\cos(\psi)$.
 - $\overline{c\psi}/N \times 10^6$ is the mean $\cos(\psi)$ per event, scaled up by 10^6 (e.g., the actual value of $\langle \cos(\psi) \rangle / N$ for the 3-D Checkerboard is 0.000419).
 - The material “EJ-Base” indicates our simulated undoped version of EJ-254, which is the base polymer (PVT) and the scintillator.
- [†]For the NuLat (3^3) experiment, IBDs were constrained to the central cube rather than distributed throughout the detector.
- [‡]For the SANDD and 2D Checkerboard experiments, directional reconstruction was performed for the azimuthal angle only.